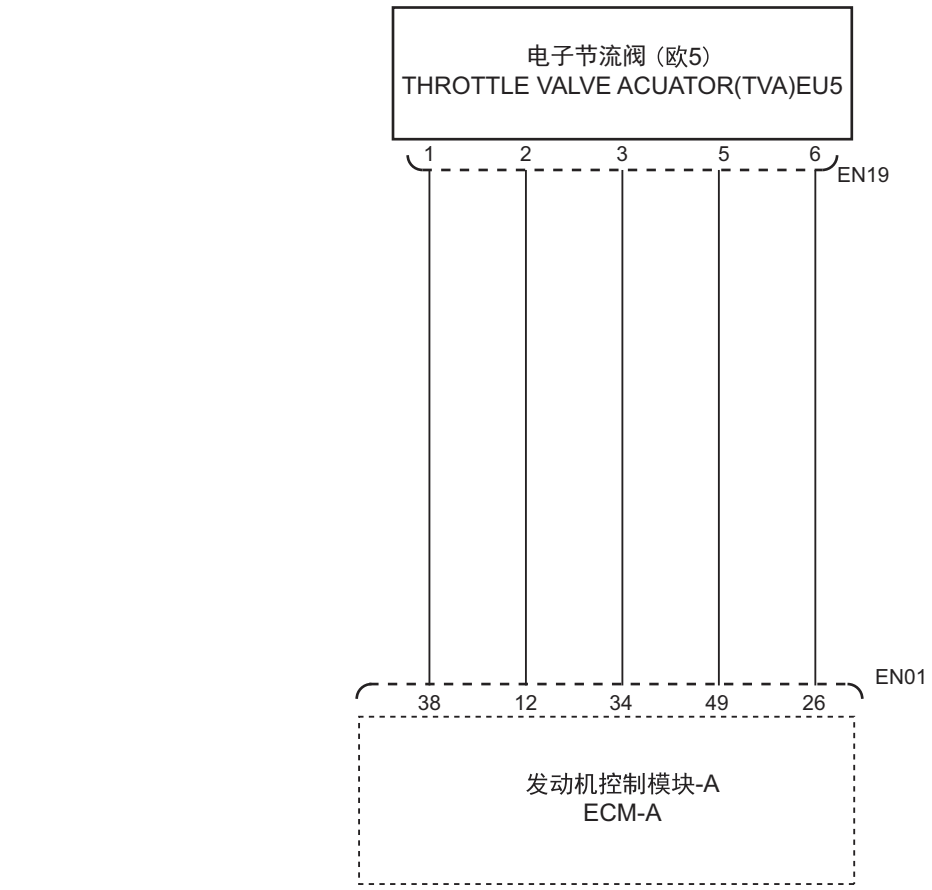
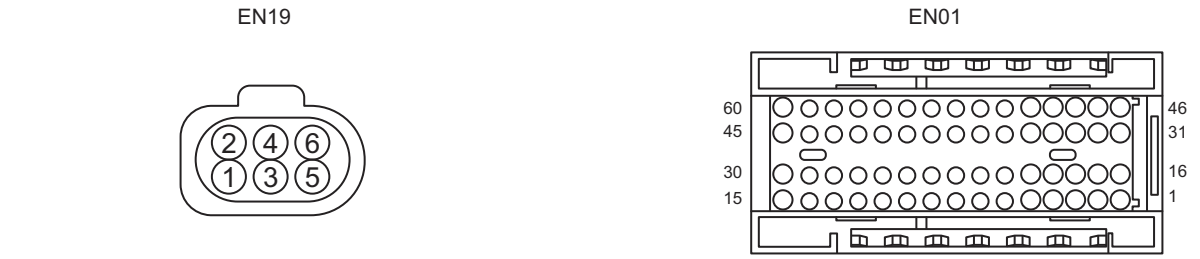
P2100

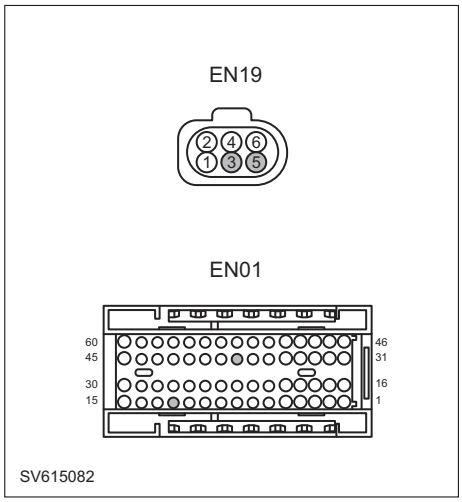
Description of fault diagnosis codes
P2100-11：Short circuit to ground on Out2 error for H-bridge of throttle valve
P2100-12：Short circuit to battery on Out2 error for H-bridge of throttle valve
P2100-13：Open load error for H-bridge of throttle valve
P2100-1D：Over current error for H-bridge of throttle valve
P2100-4B：Over temperature error for H-bridge of throttle valve
P2100-92：Temperature dependent over current error for H-bridge of throttle valve
P2100-A2：Under voltage error for H-bridge of throttle valve
P2100-F0：Short circuit to battery on Out1 error for H-bridge of throttle valve
P2100-F1：Short circuit to ground on Out1 error for H-bridge of throttle valve
P2100-F2：Short circuit over load error for H-bridge of throttle valve
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Throttle valve fault • Engine control module fault
Description of circuit diagram
The throttle is used to change the proportion of exhaust circulation indirectly by adjusting the quantity of fresh air. Generally, this valve only works when the load and speed are low. The engine control module provides 12v voltage to the drive motor and also provides grounding to the reference voltage.

Circuit diagram





SV615006

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the engine control module and the harness plug of electronic throttle valve for damage, poor contact, aging and looseness. 4 Check whether the result is normal.
		No → Repair or replace the faulted parts.
No ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for P2100?
		Yes → Reference: DTC Summary List (EU5).
No ↓		
Step 3	Check the driving circuit of electronic throttle valve (open circuit).	
		1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug of electronic throttle valve. 3 Disconnect the harness plug of engine control module. 4 Measure and check the resistance of harness plug between electronic throttle valve and engine control module. Measure circuit: the resistance between EN19 plug terminal 3 and EN01 plug terminal 12. Measure circuit: the resistance between EN19 plug terminal 5 and EN01 plug terminal 38. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		

Step 4	Check the driving circuit of electronic throttle valve (short circuit).
	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness plug of throttle valve and ground. Measure circuit: the resistance between EN19 plug terminal 3 and ground. Measure circuit: the resistance between EN19 plug terminal 5 and ground. **Standard value: ∞** 3 Measure the voltage between the harness plug of throttle valve and ground. Measure circuit: the voltage between EN19 plug terminal 3 and ground. Measure circuit: the voltage between EN19 plug terminal 5 and ground. **Standard value: 0V** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the electronic throttle valve (components).
	1 Check the resistance among the terminals of electronic throttle valve. Measure circuit: the resistance between terminal 3 and terminal 5 of electronic throttle valve connector. **Standard value: 7~12Ω** 2 Check whether the result is normal?
	No → Replace the electronic throttle valve.
Next ↓	
	Complete the check and confirm whether the system works normally nor not.